Achraf Laraje

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Profile

First-year engineering student at ENSEEIHT (Toulouse INP), Digital Sciences department, after two years of a bachelor's degree in computer science (top 4% in core subjects).

Skills

- **Embedded systems**: Arduino Due/Uno, Raspberry Pi, C/C++, sensors, serial communication, non-blocking timing.
- **Systems and networks**: C, Unix (processes, pipes), TCP/UDP client-server programming, logs, dynamic memory management.
- **Algorithms and AI**: Python, game AI, A*, Dijkstra, complexity analysis.
- **Other languages and tools**: Ada, Java, OCaml, Bash, Git, Make, Valgrind.
- **Languages**: French (native), English (B2), Spanish (intermediate).

Academic Projects

- **Electronic Escape Game: 6 puzzles, 2 Arduinos, 1 Raspberry Pi** 2025 – 2026
  Fab-Num course, team of 3, Université de Toulouse
  - **Action**: Programmed the Arduino Due for 3 puzzles (wire pairs, 45° tilt detected by accelerometer, keypad code) and the LED emergency state; restructured the main loop to read the serial port continuously; calibrated the accelerometer from its datasheet.
  - **Result**: A complete game tested end to end with the team, the Raspberry Pi coordinating both Arduinos over serial.
- **Treasure Hunt Game Server (TCP/UDP)** 2025 – 2026
  Systems and Networks course, individual project
  - **Action and result**: Designed the client-server connection, log monitoring and game synchronisation, choosing TCP or UDP for each exchange: games synchronised on a local server, with logs tracing every exchange.
- **Board Game with Artificial Intelligence (Python)** 2025 – 2026
  Project course, individual project
  - **Action and result**: Implemented the rules, move validation and pawn control, then an AI applying its own strategies against the player: graded 20/20.
- **Family Tree Manager (C)** 2024 – 2025
  C Programming course, individual project
  - **Action and result**: Built the data structures, tree traversals, insertions and deletions with explicit dynamic memory management: a tree that can be browsed and modified at any time.
- **Algorithm Visualiser (Python)** 2025, personal project
  Sorting algorithms, A and Dijkstra*
  - **Action and result**: Created a step-by-step interactive display of sorting and shortest-path algorithms on graphs and mazes, to compare their complexity visually.

Education

- **ENSEEIHT, Toulouse INP** Toulouse, France
  Engineering degree, Digital Sciences department (first year) *Sep. 2026 – Present*
- **Université de Toulouse** Toulouse, France
  Bachelor's in Computer Science (first and second years) *2024 – 2026*
  - **Grades**: 17.5/20 (year 1), 18.19/20 (year 2), top 4% in core subjects.
- **Lycée Émilie de Rodat** Toulouse, France
  French Baccalauréat, mathematics and computer science (NSI), with honours (mention Bien) *2024*

Extracurricular Activities

- **Volunteering**: homework tutor, helping pupils with their schoolwork and lessons
- **Sports**: football, running
- **Interests**: chess, photography, popular science